Bounded Instance-Support Evaluation for Finite Crop–Weed Review Queues

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Abstract

Finite crop–weed review queues must preserve distinct plants through candidate formation, role qualification, ranking, and capacity truncation. We introduce bounded instance-support evaluation, in which set coverage counts truth instances incident to qualified candidates and one-to-one support computes the maximum candidate–plant matching at a fixed queue capacity and threshold tuple. On the official WeedsGalore split, proposal commissioning increased spatial support from 0.2973 to 0.5713 and role-qualified support from 0.1928 to 0.3026. At matched burden, the commissioned-only ranker increased one-to-one support by 0.0187, whereas mismatched ranker sources reversed the effect. Mask R-CNN achieved the highest $K = 20$ point estimate (0.2159); its crossed seed-by-image contrast with U-Net was 0.0132 (95% interval, -0.0009 to 0.0300). Merge-aware evaluation revealed a 105-instance gap between set coverage and one-to-one support that was absent from AP, AR, and queue-level PQ-style summaries. Two independent operators reached 0.9421 agreement on weed reviewability ($\kappa = 0.8930$). In CWFID transfer, validation-selected thresholds exceeded external foreground responses. Image-macro plant-pixel AUROC ranged from 0.3871 to 0.6374, while threshold

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0.01 yielded spatial recall of 0.0363–1.0000 and qualified one-to-one recall up to 0.0423. Finite-queue performance therefore depends jointly on proposal formation, ranker distribution, matching contract, and external calibration.

Keywords: crop–weed segmentation, instance evaluation, review queue, matching contract, quality assurance, precision agriculture

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1. Introduction

Site-specific weed management links sensing, plant localisation, crop–weed discrimination, and intervention [1, 2]. Field-scale mapping protocols must align imaging, ground validation, and the management question [3]. The same alignment is needed when predicted regions enter an offline review queue. A high pixel score alone leaves unresolved how many distinct plants remain available when inspection capacity is finite.

Standard instance metrics answer complementary questions. COCO-style average precision and average recall evaluate detection and mask quality across intersection-over-union thresholds and detection limits [4]. Panoptic Quality combines recognition and segmentation quality after one-to-one matching [5]. Recent analyses show that average precision can obscure duplicate predictions and can change abruptly with matching thresholds [6, 7]. Finite agricultural review adds two quantities to this metric family: an explicit inspection capacity and the stage at which each plant leaves the candidate-to-queue path.

Finite review introduces several loss mechanisms. A plant can be absent from the proposal pool, contained in a mixed crop–weed region, merged with neighbours, ranked below the cut-off, or duplicated by several candidates. Object-proposal research relates recall to overlap and proposal count [8]. Precision-at-the-top methods address ranking under restricted capacity [9]. Agricultural queue evaluation therefore needs a contract that connects proposal support, role evidence, multiplicity, ranking, and the number of records available for inspection.

The queue studied here supports offline annotation quality assurance and model review. An operator can retain a region, reject it, or escalate an ambiguous mixture before any downstream action. This record-oriented task complements active learning, which selects observations for annotation or retraining [10, 11]; selective prediction, which abstains from automated outputs [12, 13]; and learning to defer, which assigns cases between a model and an expert [14, 15].

We present a bounded instance-support evaluation for finite crop–weed review queues. It has four contributions. First, it separates spatial, role-qualified, set-coverage, and one-to-one endpoints. Second, it tests proposal effects across 27 matching contracts and four ranker-training schemes. Third, it combines seed and test-image variation and compares the queue endpoint with standard AP, AR, and PQ-style metrics. Fourth, it connects geometric support to independent operator review and a locked cross-dataset transfer. Together, these components establish a reproducible evaluation framework for agricultural review queues.

2. Materials and methods

2.1. Review task and endpoint bounds

Each review item contained one RGB tile, one highlighted connected region, and a local RGB view. The intended output was one quality-assurance record: retain, reject, or escalate. We fixed $K = 20$ as a common experimental capacity across systems.

One-to-one support reflects the record contract. For a fixed queue capacity K and matching threshold tuple, a selected region can contribute at most one edge to the maximum-cardinality candidate–instance matching. Set coverage is the number of truth instances incident to at least one edge under that same tuple. The first endpoint is sensitive to duplicate and merged regions; the second retains every qualified membership and is therefore merge-tolerant. Their ordering characterises the thresholded bipartite graph, while the operator protocol independently measures human reviewability.

Figure 1 separates inference from offline scoring and places the endpoints within the evidence chronology. Target truth can enter declared training or validation roles. Test masks and instance identifiers enter only offline scoring after proposal generation, ranking, and queue allocation. Taxonomic identity, agricultural role, and management morphotype remain distinct axes. The experiments evaluate the crop–weed role axis.

2.2. Datasets, splits, and evidence classes

The source specialists used public SugarBeets2016 RGB images and masks [16]. Target development used 156 annotated 600×600 pixel WeedsGalore tiles from one maize field [17]. The official allocation contains 104 training,

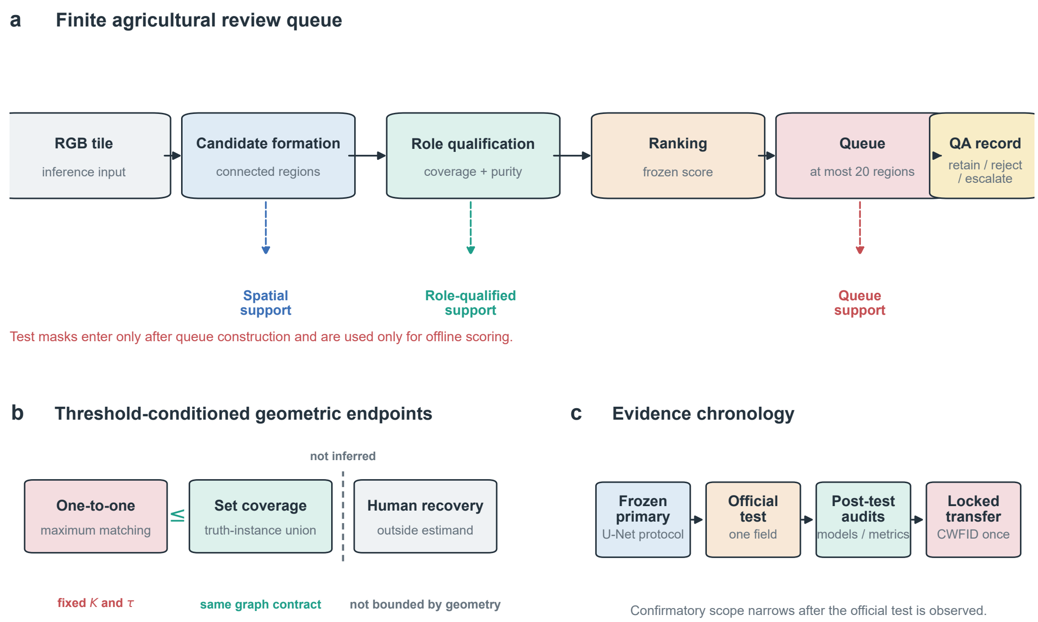

Figure 1: Evaluation contract and evidence chronology. (a) RGB inference produces a finite queue before test masks enter offline scoring. (b) At fixed K and matching thresholds, one-to-one support is the maximum candidate–instance matching cardinality and set coverage is the incident truth-instance union; human reviewability is measured separately. (c) The frozen U-Net analysis preceded official-test access, followed by model, ranker, contract, and standard-metric experiments. CWFID formed a separately locked external transfer.

26 validation, and 26 test tiles. Raster enumeration yielded 2,168 crop and 10,029 weed instances. The official test masks contained 256 crop and 1,712 weed instances.

The test split represents four acquisition dates (Table 1). The first three dates contribute eight tiles each and the last date contributes two. Image bootstrap quantifies within-split variation, while date-block resampling quantifies variation across the four observed acquisition dates.

Analyses were organised by evidence chronology. The frozen primary experiment transferred five validation-selected U-Net checkpoints to CWFID. The matched-burden, strong-model, ranker-source, contract-grid, merge-aware, operator, and standard-metric experiments used the official WeedsGalore test split. Continuous-score and threshold-response analyses were performed after the locked CWFID endpoint. This chronology is retained in the run registry

Table 1: WeedsGalore official-test denominators by acquisition date.

Acquisition date	RGB tiles	Weed instances	Crop instances
25 May 2023	8	392	122
30 May 2023	8	440	55
6 June 2023	8	726	68
15 June 2023	2	154	11
Total	26	1,712	256

and result tables.

Cross-dataset evaluation used the official CWFID v1.0 release [18]. Its 60 RGB images and public annotations were fixed at source commit `176612f3`. The protocol fixed preprocessing, five existing U-Net checkpoints, validation-selected settings, $K = 20$, and endpoints before data access. All 242 upstream blobs were verified before one recorded opening. Weed truth units were defined as eight-connected annotation components of at least 16 pixels. Model fitting and selection used no CWFID labels.

2.3. Supervision ledger

Table 2 reports the supervision burden. Proposal generation is image-only during inference, while operating-point commissioning uses target-training masks and candidate role labels derived from dense masks. The 300-candidate ablation samples training candidates while retaining all 2,436 labelled validation candidates.

2.4. Candidate formation, rankers, and model families

Two ResNet-18 U-Net-style source specialists were initialised with ImageNet-1K weights [19]. One estimated plant foreground and the other estimated crop and weed probabilities. Thresholding foreground probability at T, applying eight-connectivity, and removing components below A pixels defined setting (T, A). Full training details appear in the supplement.

The original operating point was $(0.95, 32)$. Proposal operating-point commissioning searched 12 settings on the 104 official-training tiles. Feasible settings produced at most 100 candidates per tile on average, preserving a proposal pool wider than the downstream $K = 20$ review capacity. Training role-qualified proposal recall was maximised, followed by lower burden, higher threshold, and larger area. The selected setting was $(0.70, 32)$.

Table 2: Supervision and data roles. Test truth is restricted to offline scoring.

System or stage	Training supervision	Selection supervision	Test use
Source specialists	7,050 SugarBeets RGB masks	901 source masks for source assessment	RGB inference only
Proposal commissioning	104 target dense masks	Training-mask objective and burden constraint	Offline scoring
Source-transfer rankers	9,887 target-training candidate labels	2,436 validation candidate labels	Queue formed before scoring
U-Net and DeepLabV3+	104 target RGB dense masks	26 validation RGB dense masks	Queue formed before scoring
Mask R-CNN	104 target RGB semantic and instance masks	26 validation semantic and instance masks	Queue formed before scoring
CWFID transfer	No external labels	No external labels	60 annotations for locked scoring only
Two-operator review	No model training	Frozen 518-region queue and common instructions	Aggregate technical-review analysis

Five candidate rankers used source weed score, geometry, role-plus-local features, frozen DINOv2 features, or component-masked DINOv2 features [20]. Logistic regularisation was selected on official validation. The burden-matched proposal experiment tested four training schemes: commissioned-only, original-only, the union of both pools, and reciprocal cross-pool transfer. The reciprocal scheme trained on one pool and applied that ranker to the other pool.

Target semantic models comprised a ResNet-18 U-Net and DeepLabV3+ with a ResNet-50 encoder [21]. The instance model was Mask R-CNN with a ResNet-50 FPN backbone [22]. U-Net used five training seeds. DeepLabV3+ and Mask R-CNN used three. Validation selected each checkpoint and component policy. All model families used the same $K = 20$ endpoint. Training used one NVIDIA H200 accelerator per seed.

2.5. Support, matching, and standard metrics

For truth instance I_j and candidate C_i, instance coverage is

$$\mathrm{cov}(I_j, C_i) = \frac{|I_j \cap C_i|}{|I_j|}. \tag{1}$$

Spatial support requires at least one candidate with coverage of 0.50. Role-qualified support additionally requires 0.50 labelled-pixel coverage and 0.90 same-role purity. These values form the primary matching contract. A $3\times3\times3$ grid used instance coverage, labelled coverage, and purity values of 0.25, 0.50, and 0.75, with 0.90 replacing 0.75 at the highest purity level.

For a selected queue Q_K and primary threshold tuple $\boldsymbol{\tau} = (0.50, 0.50, 0.90)$, let $E_{\boldsymbol{\tau}}$ contain every candidate–weed-instance pair meeting the instance-coverage, labelled-coverage, and same-role-purity thresholds. Set coverage is $|\{I_j : \exists C_i \in Q_K, (C_i, I_j) \in E_{\boldsymbol{\tau}}\}|$. One-to-one support is the maximum-cardinality matching on the same thresholded graph. Candidate and instance identifiers resolve ties. The difference between these numerators is the merge-capacity gap after duplicate memberships are removed. Candidate precision is the fraction of selected regions satisfying the role-qualified weed contract. Every reported endpoint is indexed by K and $\boldsymbol{\tau}$; changing either changes the estimand.

Crop exposure is the fraction of tiles with a queued region overlapping crop truth. The analysis uses any positive overlap and candidate-area overlap thresholds of 1% and 10%. Standard instance metrics include mAP over IoU 0.50–0.95, AP50, AP75, AR100, and queue AR20. A PQ-style summary covers the predicted-weed subset of the $K = 20$ queue; a complete panoptic partition would additionally require background and crop segments.

2.6. Matched burden and uncertainty

Commissioning changes candidate support and burden. A common per-tile capacity used

$$k_i = \min(20, n_i^{\text{original}}, n_i^{\text{commissioned}}) \tag{2}$$

for test tile i. Both settings contributed 471 candidates. Ranker-free proposal-set ceilings and per-tile $K = 20$ one-to-one oracles separate formation capacity from achieved ranking.

The primary model comparison uses 20,000 crossed bootstrap replicates. Each replicate samples a training seed within each model and resamples the 26 test images with replacement. The two prespecified one-to-one contrasts are DeepLabV3+ minus U-Net and Mask R-CNN minus U-Net. We report 95% intervals and Bonferroni-adjusted familywise 97.5% intervals. A date-block plus seed bootstrap quantifies sensitivity across the four observed acquisition dates.

2.7. Two-operator technical review

Two different human operators independently reviewed the frozen 518-region U-Net queue using common written instructions and 12 practice items from validation data. Formal reviews were completed independently, with peer responses, automated labels, external searches, acquisition date, tile identity, candidate identifiers, scores, ranks, model identity, and mask-derived truth unavailable to either operator.

The operators recorded primary content, weed reviewability, crop presence, localisation usability, and five-point confidence; timing lay outside the measurement set. Exact agreement, Cohen's κ, Gwet's AC1, and scene-cluster intervals were computed. Linear and quadratic weighted κ were added for ordered fields. Human consensus and mask qualification were compared in a three-by-two contingency table over all 518 regions. Binary precision, recall, F1, Jaccard, and accuracy used regions with determinate consensus (both "yes" or both "no"), while disputed or uncertain regions remained in the contingency table.

3. Results

3.1. Commissioning expanded raw support, while queued effects depended on ranker training

The original setting produced 802 test candidates, or 30.85 per tile. The commissioned setting produced 2,351, or 90.42 per tile. Spatial proposal recall increased from 0.2973 to 0.5713. Role-qualified proposal recall increased from 0.1928 to 0.3026. These ranker-free changes establish greater raw geometric support.

At the common 471-candidate burden, the commissioned-only ranker increased spatial recall by 0.0818, set coverage by 0.0374, and one-to-one recall by 0.0187. Precision decreased by 0.0679. The primary-contract one-to-one change was 0.0210 with the union ranker, -0.0514 with the original-only ranker, and -0.0625 under reciprocal transfer (Table 3). The change in sign identifies ranker-source alignment as a determinant of the queued effect.

The same dependence appeared across 27 contracts (Figure 2). Spatial support increased in all contracts for commissioned-only and union rankers. One-to-one support increased in 15 and 16 contracts, respectively. It decreased in every contract for original-only and reciprocal rankers. Under the primary contract, the original and commissioned proposal-set ceilings were 0.1928

Table 3: Commissioned-minus-original effects at the common 471-candidate burden under the primary matching contract.

Ranker training pool	Precision	Spatial	Set coverage	One-to-one
Commissioned only	−0.0679	+0.0818	+0.0374	+0.0187
Original only	−0.3503	−0.1268	−0.0695	−0.0514
Union	−0.0701	+0.0847	+0.0380	+0.0210
Reciprocal cross-pool	−0.3482	−0.1460	−0.0847	−0.0625

and 0.3026. Their per-tile $K = 20$ oracles were 0.1396 and 0.1618. Achieved one-to-one support was 0.0970 and 0.1157.

3.2. Set coverage exposed a merge-capacity gap

The 518-region U-Net queue contained 170 regions with no qualified weed instance, 286 with one, and 62 with two or more. One merged region covered nine qualified instances. The union of qualified memberships contained 453 distinct instances. One-to-one matching retained 348. Set coverage was therefore 0.2646, whereas one-to-one support was 0.2033. The 105-instance difference arose from merged capacity; no duplicate membership remained after the set union.

Among 301 consensus-reviewable regions, set coverage was 0.2021 and one-to-one support was 0.1454 under the primary threshold tuple. Their 97-instance graph gap again reflected merges. Reviewability increased with geometric multiplicity: 0.3059 for no qualified instance, 0.6818 for one, 0.8140 for two, and 1.0000 for three or more. This association links geometric multiplicity to region-level reviewability, while recoverable plant counts remain a separate measurement task.

3.3. Standard metrics and crossed uncertainty changed the model comparison

Mask R-CNN had the highest one-to-one point estimate at $K = 20$: 0.2159 ± 0.0038 across three seeds. U-Net reached 0.2027 ± 0.0057 across five seeds, and DeepLabV3+ reached 0.1924 ± 0.0032 across three. The crossed seed-by-image contrast was +0.0132 for Mask R-CNN versus U-Net, with a 95% interval of −0.0009 to 0.0300 and a familywise interval of −0.0027 to 0.0329. DeepLabV3+ minus U-Net was −0.0103, with a 95% interval of −0.0237 to 0.0026. Both 95% intervals crossed zero.

The date-block plus seed interval was 0.0023 to 0.0341 for Mask R-CNN minus U-Net and −0.0244 to 0.0133 for DeepLabV3+ minus U-Net. Across

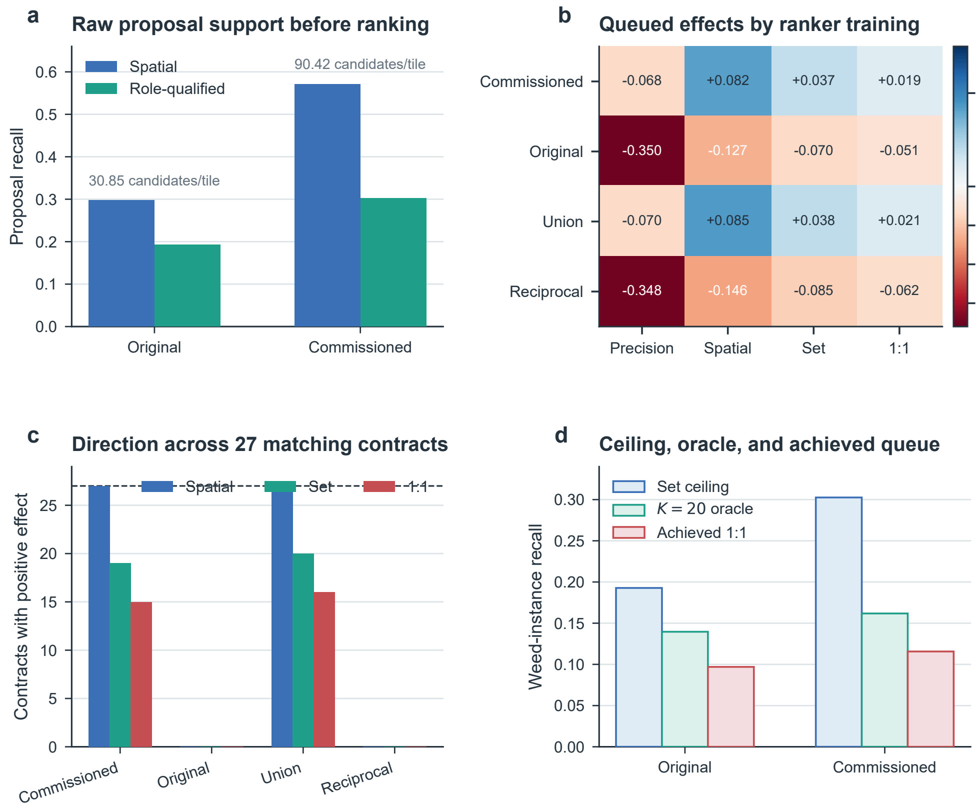

Figure 2: Proposal and ranker sensitivity. (a) Raw spatial and role-qualified proposal support before ranking. (b) Commissioned-minus-original effects under four ranker-training schemes. (c) Number of positive effects across 27 matching contracts. (d) Ranker-free set ceiling, per-tile $K = 20$ oracle, and achieved one-to-one support for the commissioned-only ranker.

the observed dates, point estimates and standard metrics consistently favoured Mask R-CNN, while the crossed image-level interval retained uncertainty around its contrast with U-Net.

Standard metrics on the same predictions also favoured Mask R-CNN (Table 4). Its mAP was 0.1064 and AP50 was 0.3189. U-Net reached 0.0853 and 0.2616, respectively. At the queue limit, Mask R-CNN reached AR20@50 of 0.1861 and a PQ-style score of 0.1936. The matching-contract grid was less uniform: Mask R-CNN ranked first in 21 of 27 contracts, while U-Net ranked first in six. U-Net led when instance coverage was 0.75 with labelled coverage of 0.25 or 0.50.

Table 4: WeedsGalore instance metrics on the same predicted-weed regions. Each cell reports the seed mean with the sample SD in parentheses.

Model	Seeds	One-to-one $K = 20$	mAP	AP50	AR100@50	Queue AR20@50
U-Net	5	0.2027 (0.0057)	0.0853 (0.0037)	0.2616 (0.0157)	0.3412 (0.0207)	0.1672 (0.0070)
DeepLabV3+	3	0.1924 (0.0032)	0.0499 (0.0026)	0.1937 (0.0060)	0.2950 (0.0081)	0.1456 (0.0045)
Mask R-CNN	3	0.2159 (0.0038)	0.1064 (0.0082)	0.3189 (0.0248)	0.3904 (0.0266)	0.1861 (0.0079)

Representative queue structures explain why the endpoints diverge (Figure 4). A split plant occupies several reviews, whereas a merged candidate supports several truth instances through one record. Background regions and crop–weed mixtures consume capacity outside the primary role contract.

3.4. Operators agreed on categories, but human and geometric criteria differed

Both operators completed all 518 regions. Weed-reviewability agreement was 0.9421, Cohen's κ was 0.8930, and Gwet's AC1 was 0.9206. The scene-bootstrap interval for AC1 was 0.8918 to 0.9483. Primary content, crop presence, and localisation AC1 values were 0.9355, 0.9341, and 0.9243.

Confidence matched exactly for 0.9440 of items. Linear weighted κ was 0.9366 and quadratic weighted κ was 0.9589. Mean confidence was 3.523 for one operator and 3.510 for the other. Both medians were 4, with interquartile ranges of 3–4. Among 445 determinate localisation pairs, exact agreement was 0.9955 and quadratic weighted κ was 0.9966.

The operators reached consensus that 301 regions were reviewable and 143 were not. Seventy-four were disputed or contained uncertainty. Mask qualification agreed with consensus reviewability on 249 positive and 76 negative regions. It produced 67 mask-positive/human-negative and 52 mask-negative/human-positive regions. Thirty-nine unresolved regions were mask-positive and 35 were mask-negative. On the 444 determinate-consensus regions (85.71% of all items), mask precision was 0.7880, recall was 0.8272, F1 was 0.8071, Jaccard was 0.6766, and accuracy was 0.7320. Human reviewability and fixed geometry are related but non-equivalent evidence.

3.5. External transfer exposed threshold-sensitive plant evidence

The locked CWFID evaluation contained 60 images and 331 weed-component proxies. For all five U-Net checkpoints, the maximum external

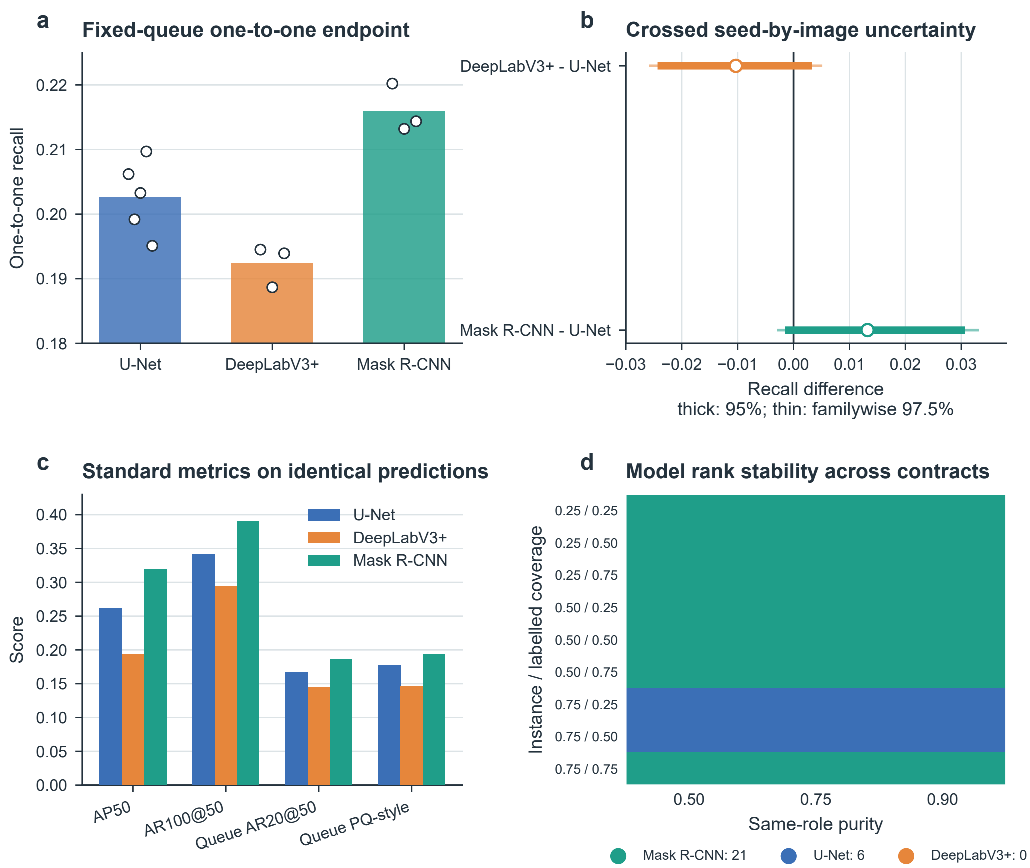

Figure 3: Model comparison on the 26 WeedsGalore test tiles. (a) Seed-level one-to-one support. (b) Crossed seed-by-image contrasts with U-Net. Thick lines show 95% intervals; thin lines show Bonferroni-adjusted familywise 97.5% intervals. (c) Standard metrics on identical predictions. (d) Best one-to-one model across 27 matching contracts.

foreground response remained below the validation-selected threshold, producing an empty connected-component queue. Continuous plant-pixel discrimination varied substantially across checkpoints: image-macro AUROC ranged from 0.3871 to 0.6374 and AP from 0.0438 to 0.0715 (Figure 5a).

Image-macro foreground q99 values ranged from 0.0118 to 0.0498 and image-macro maxima from 0.0384 to 0.2205, compared with validation-selected thresholds of 0.30–0.70 (Figure 5b). The resulting score-scale separation explains why connected-component extraction remained inactive at the registered operating points.

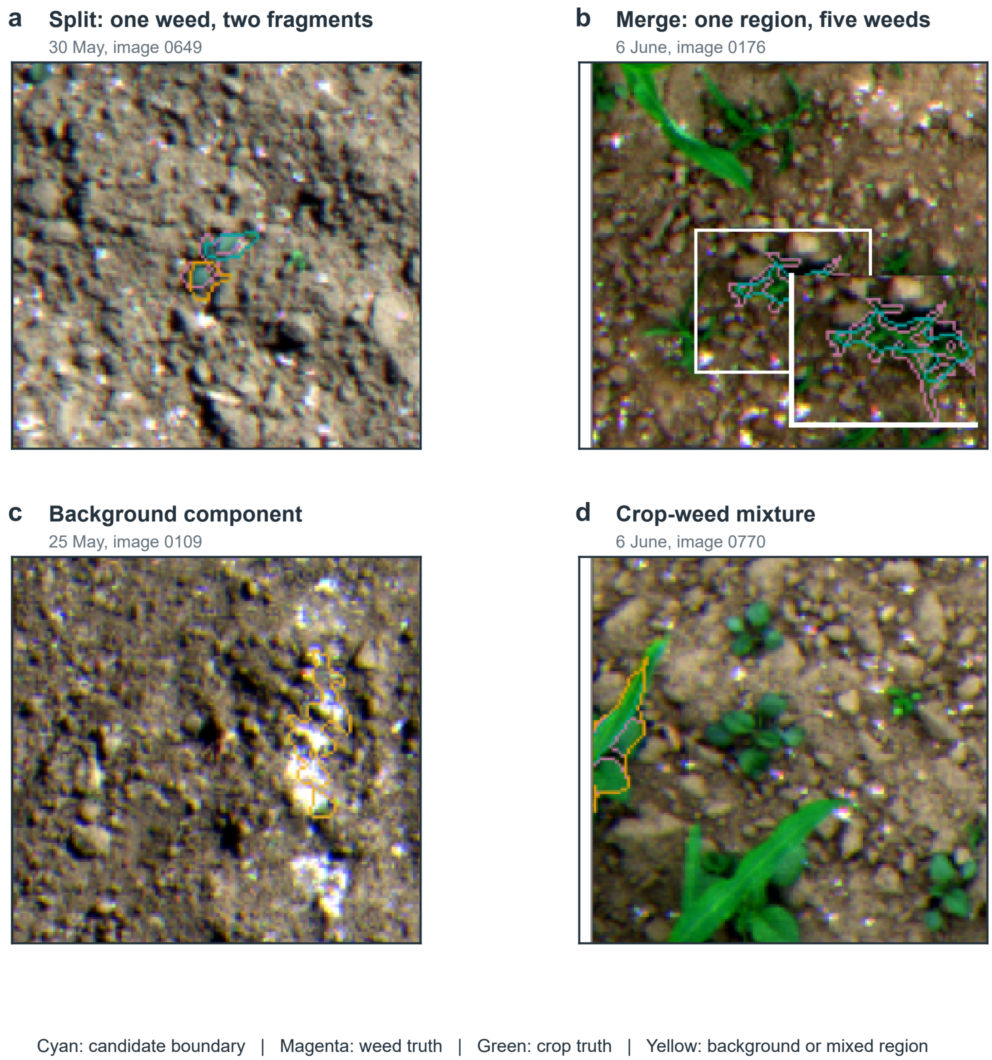

Figure 4: Representative proposal structures from the official WeedsGalore test split. (a) A weed split into two candidates. (b) One candidate covering five weed instances, with a local enlargement. (c) A background component. (d) A crop–weed mixture. Truth contours provide the offline spatial reference.

At threshold 0.01, spatial recall ranged from 0.0363 to 1.0000, candidate precision from 0 to 0.3520, and crop-exposure frequency from 0.25 to 1.00 (Figure 5c). Role-qualified one-to-one recall was positive for one checkpoint and reached 0.0423. Checkpoint 4 combined spatial recall of 1.0000 with crop

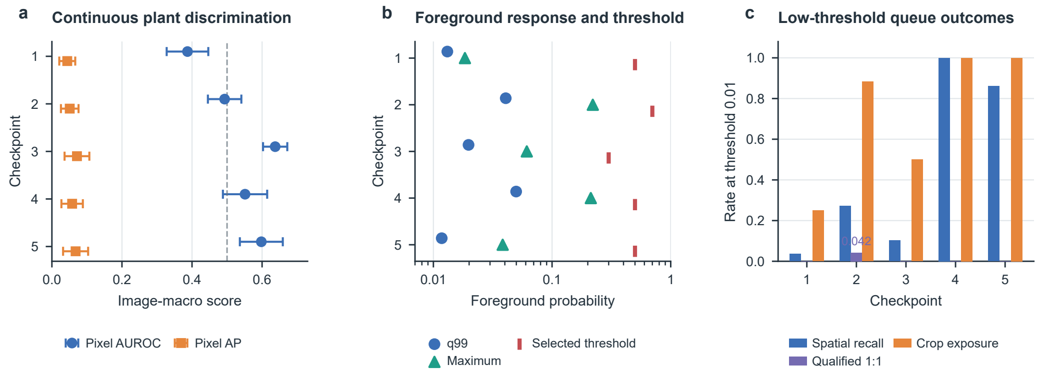

Figure 5: External CWFID transfer across five U-Net checkpoints. (a) Image-macro plant-pixel AUROC and AP; error bars show the sample SD across 60 images. (b) Image-macro foreground q99 and maximum probabilities relative to the validation-selected threshold. (c) Spatial recall, role-qualified one-to-one recall, and crop-exposure image frequency at threshold 0.01. Checkpoints follow the registered seed order.

exposure of 1.0000 and qualified recall of 0, localising transfer loss to both score scale and crop–weed role qualification.

4. Discussion

The evaluation contract extends candidate discrimination and standard instance scores with an explicit agricultural review capacity. AP, AR, and PQ-style metrics measure localisation, segmentation, ranking, and duplicate errors under established conventions. Set coverage and one-to-one support add a stage-wise account of proposal, role, merge, ranking, and truncation effects.

The merge-aware analysis distinguishes two graph quantities under the primary threshold tuple and $K = 20$. One-to-one support is the maximum number of candidate–plant pairs that can be assigned without reusing either node. Set coverage is the truth-instance union over all qualified edges and therefore retains multiple memberships of a merged candidate. Their 105-instance difference quantifies merge capacity in this thresholded graph. The operator experiment complements this geometry with region-level reviewability, separating graph structure from human interpretation.

Proposal operating-point commissioning improved ranker-free spatial and role-qualified support. Its queued benefit depended on which proposal distribution trained the ranker. Commissioned-only and union training produced positive median effects, whereas original-only and reciprocal transfer reversed them. This result distinguishes a proposal ceiling from an achieved queue. A larger candidate pool can contain more threshold-qualified support while remaining poorly ordered by a mismatched ranker.

Mask R-CNN led 21 of 27 matching contracts and all reported standard metrics. Its one-to-one point estimate exceeded U-Net by 0.0132, with a crossed seed-by-image 95% interval of -0.0009 to 0.0300. Date-block resampling shifted the interval to 0.0023–0.0341. The contrast therefore combines a consistent ranking across metrics with sensitivity to the chosen resampling unit.

The two operators applied the review categories consistently, with weed-reviewability agreement of 0.9421 and AC1 of 0.9206. The 74 disputed or uncertain regions define the escalation branch of the workflow. On determinate consensus items, the human–mask Jaccard index of 0.6766 shows that geometric qualification and human reviewability capture related but distinct properties. Operator-population variation, plant-count extraction, review speed, and labour economics remain separate study endpoints.

The CWFID experiment reveals a two-stage transfer pattern. First, external foreground responses fell below every validation-selected threshold. Second, threshold relaxation recovered spatial coverage for several checkpoints but yielded little role-qualified one-to-one support and frequent crop exposure. This separation identifies score-scale transfer and agricultural role qualification as distinct bottlenecks. The registered external experiment used the five U-Net checkpoints available before CWFID opening; evaluation of the later Mask R-CNN family is a natural next external comparison.

For annotation quality assurance, the contract links four design choices—candidate source, role criterion, review budget, and escalation rule—to proposal ceilings, achieved support, crop exposure, and standard AP/AR. Ranker-source and matching-contract grids show whether improvements persist across training distributions and overlap criteria. External transfer then tests score scale and role qualification under a new acquisition domain. Timing, plant-count extraction, and cost can be incorporated as additional operational endpoints.

4.1. Limitations

The evidence spans one WeedsGalore field with four acquisition dates and one external CWFID collection. Image and date-block intervals quantify variation within these observed acquisitions. CWFID changes crop, sensor geometry, scale, and annotation contract, with connected components serving as plant proxies. The operator experiment includes two individuals and region-level decisions; timing and recoverable-plant counts were outside its measurement design. The fixed $K = 20$ capacity provides a common experimental budget, while economic optimisation requires task-specific labour and field-cost data. External comparison of all three model families would further resolve architecture-specific transfer effects.

5. Conclusion

Bounded instance-support evaluation connects crop–weed predictions to a finite offline review queue. On WeedsGalore, commissioning nearly doubled spatial proposal support, while matched-burden one-to-one gains depended on ranker-source alignment. Merge-aware matching revealed a 105-instance capacity gap, and Mask R-CNN achieved the highest $K = 20$ point estimate across the tested models. Independent operators applied the review categories consistently, while their consensus remained distinct from mask qualification. On CWFID, foreground responses shifted below validation-selected thresholds; threshold relaxation recovered spatial coverage but little role-qualified one-to-one support. These results establish proposal formation, ranker distribution, matching contract, and external score calibration as separate determinants of agricultural review-queue performance.

Data availability

SugarBeets2016, WeedsGalore, and CWFID are public datasets available from their cited sources. The reproducibility release records source URLs, access dates, release identifiers, licence information, versions, and integrity hashes. Public imagery remains with its original repositories. Derived aggregate tables, matching outputs, and verification manifests accompany the code release.

Code availability

Versioned reproducibility materials are maintained at `https://github.com/Fytap/agrispec-vlm-reproducibility`. The manuscript-linked v1.3.1 evidence layer is fixed at the immutable commit `https://github.com/Fytap/agrispec-vlm-reproducibility/tree/6930f7b48995054095dbaa` eebef4f155029c107a/reproducibility/v1_3_1. The compact verifier replays archived tables, figures, and numerical invariants. Full end-to-end reproduction uses the supplied training and evaluation scripts, public data instructions, environment specification, checkpoint hashes, model acquisition records, and complete run logs.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the authors used AI-assisted tools for code review and language review, including checks of sentence clarity, grammar, and wording. The authors reviewed and edited all affected content and take full responsibility for the content of the publication.

Ethics statement. No external participants were recruited and no personal data were collected; human-participant ethics approval and informed consent were not applicable.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

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